

FIIB
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Advance Operations Management [OP16]

Vehicle Routing Optimization for Last-Mile Delivery

Operations Research Project — Scheduling & Sequencing

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Learning Team - 5

Problem Statement & Objective

Problem Statement

- Logistics company serving 12 delivery locations
- Single warehouse (Depot W)
- Need to minimize total delivery distance
- Improve last-mile delivery efficiency

169 Binary Variables
13 × 13 Decision Matrix
Excel Solver (Simplex LP)

Objectives

- Minimize total delivery distance
- Optimize delivery routes
- Improve vehicle utilization
- Reduce operational cost
- Improve delivery efficiency

Dataset & Methodology

43,739 Delivery Records

Traffic • Weather • Vehicle • Area

Methodology

Construct 13-node network (12 stops + warehouse)

Create distance matrix (Euclidean distance)

Create binary decision matrix

Apply Excel Solver

Generate optimal delivery routes

Amazon Delivery Operations Dataset (Kaggle)

- **43,739 delivery records**
- **Traffic conditions**
- **Weather conditions**
- **Delivery time**
- **Vehicle type**



Mathematical Model & Solver Setup



Mathematical Model
Decision Variable:
Xij = 1 if route selected
Xij = 0 otherwise

Objective Function
Minimize:
Total Distance = Σ Σ Dij × Xij

Constraints
Each location visited once
Each location departed once
Binary decision variables
No self-loops

Tool Used
Microsoft Excel Solver
Method: Simplex LP
Decision Variables: 169

Results & Optimized Routes

Vehicle	Route
Vehicle 1	W → S09 → S10 → W
Vehicle 2	S01 → S02
Vehicle 3	S03 → S04
Vehicle 4	S05 → S06
Vehicle 5	S07 → S08
Vehicle 6	S11 → S12

Key Results

Total Distance: 268.03 units

12 locations served

6 optimized delivery routes

Solver Time: 0.141 seconds

34 Packages Delivered



Business Insights & Conclusion

Business Insights

Parallel routing improves delivery speed

Cluster-based routing reduces cost

Better vehicle utilization

Improved last-mile efficiency

10-15% Cost Reduction

46% Traffic Impact

30% Weather Impact



Conclusion

Vehicle Routing Optimization improves delivery efficiency

Excel Solver successfully optimized routes

Model applicable to real-world logistics operations

Scalable to large delivery networks